

XAI for U 2025: 2nd International Workshop on Explainable AI for Ubiquitous, Pervasive and Wearable Computing

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Abstract

The workshop *XAI for U* aims to address the critical need for transparency in Artificial Intelligence (AI) systems that are increasingly integrated into our daily lives through mobile systems, wearables, and smart environments. Despite rapid advances in AI and machine learning, many of these systems remain opaque, making it difficult for users, designers, developers, and stakeholders to verify their reliability, fairness, and correctness. This lack of transparency can hinder trust, informed decision-making, and broader adoption. This workshop focuses on the pressing need to enable Explainable AI (XAI) tools tailored for ubiquitous, pervasive, and wearable computing, and highlights the unique challenges associated with this domain, such as generating explanations for time-series and multimodal data, interpreting interconnected machine learning components, and delivering user-centered explanations. It aims to foster interdisciplinary collaboration among researchers across related domains, share recent advancements, address open challenges, and propose future research directions to improve the development and applicability of XAI in ubiquitous, pervasive, and wearable computing. Ultimately, the workshop seeks to enhance user trust, understanding, interaction, and adoption, ensuring that AI-driven solutions are not only more explainable but also better aligned with ethical standards and user expectations.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

XAI, Explainable AI, Ubiquitous Computing, Wearable Computing, User Studies

ACM Reference Format:

Mor Vered, Teena Hassan, Mandani Ntekeouli, Sang Won Bae, and Martin Gjoreski. 2025. XAI for U 2025: 2nd International Workshop on Explainable

AI for Ubiquitous, Pervasive and Wearable Computing. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3750557>

1 Background and Motivation

1.1 Introduction to XAI for Ubiquitous Computing

Ubiquitous and wearable computing, exemplified by mobile systems, wearable technologies, and intelligent environments, utilizes Artificial Intelligence (AI) to develop advanced solutions that improve various aspects of our lives, from healthcare and sports performance to overall quality of life. Furthermore, ubiquitous technologies allow users to interact with AI-enabled agents through a wide array of devices and everyday objects, such as computers, smartphones, smart glasses, smart mattresses, and smart home appliances like refrigerators. In the age of ubiquitous AI-based systems that deeply integrate into human lives, it is critical that users understand how these systems operate and make decisions. Unfortunately, today's groundbreaking AI methods are black boxes (i.e., the decision model and the decision process are not understandable). This opacity unveils a critical need for Explainable AI (XAI) tools, through which all stakeholders (e.g., users, designers, developers, employers, clinicians, policymakers) can verify the correctness and reliability of decisions made by AI-driven ubiquitous solutions. As AI continues to integrate into our lives, making its operations transparent through XAI is not just a preference, but a necessity.

The development of XAI began in the 1980s with systems like XPLAIN and EMYCIN [17, 18], which aimed to explain expert systems. By the late 1990s, the focus expanded to explainers for Bayesian networks [1, 11] and recommender systems [6, 9]. In recent years, the need for XAI has grown with the rise of complex AI approaches, which are increasingly applied in the context of ubiquitous and wearable computing. For example, studies have explored XAI for human activity recognition models [3, 4, 10]; Sharma et al. [16] utilized XAI in their machine learning (ML) solution for psychophysiological arousal in children who stutter; Rahman et al. designed an interpretable framework to empower people with Parkinson's disease [14]; Das Antar et al. explored quantitative and qualitative model explanations to understand models to predict



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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3750557>

physical functioning of people with Multiple Sclerosis [2]; and Bae et al. demonstrated the feasibility of real-world XAI applications by developing interpretable models to detect acute marijuana intoxication [5]. While most existing approaches focus on generating post-hoc explanations for single decisions or actions of an AI-based system, prior work has proposed treating explanations as an interactive process through which understanding of the explanandum is co-constructed by a machine explainer and a human explainee [15].

In addition to the increasing trend of XAI studies, UbiComp studies have shown that the availability of explanations positively influenced the willingness of users to adopt personal sensing technologies [12]. Furthermore, two studies have shown that end users often struggle to understand even high-level summaries of Fitbit and similar sensing platforms [7, 19]. However, despite the clear need for XAI in the UbiComp community, there has not yet been a dedicated XAI workshop at UbiComp.

1.2 XAI for U at UbiComp 2025

XAI has become a particularly hot topic in recent years: the European Union engrossed the AI Act, which mandates human oversight and expands the range of tasks in which XAI can be applied [13]. DARPA launched the XAI initiative; HORIZON Europe regularly issues XAI-related project calls; and the Swiss Hasler foundation opened the Responsible AI Funding Program. Additionally, major AI conferences, such as IJCAI, AAAI, and NeurIPS, have featured workshops dedicated tracks to XAI, reflecting its growing importance in the research community. The International ACM Conference on Fairness, Accountability, and Transparency (FAccT) is also largely focused on XAI. While these venues emphasize algorithmic advancements, the unique challenges of applying XAI within the context of ubiquitous and wearable computing are neither adequately addressed nor sufficiently represented. Case in point, last year's UbiComp conference included no papers specifically focused on XAI for wearable computing [20].

In particular, XAI in ubiquitous and wearable computing faces several challenges. Research has largely focused on image and text data, with limited attention to time-series data from wearable sensors and IoT devices. Many studies emphasize single-source models, neglecting inherently multimodal nature of these systems, while existing XAI tools often explain individual models rather than interconnected machine learning pipelines. User studies evaluating XAI remain scarce, and most solutions are static, failing to adapt to dynamic environments. Limited user understanding of platforms like Fitbit highlights the need for end-to-end explanations, while privacy and ethical concerns require careful integration. XAI deployment on wearable devices is underexplored, and solutions must support personalized, context-aware explanations to meet diverse user needs. Additionally, the role of explanations in influencing user trust—whether through over- or under-reliance on decision-support tools—requires further investigation to ensure meaningful and responsible engagement.

2 Objectives

The *XAI for U* Workshop aims to advance the development and practical applicability of explainable AI (XAI) in the domain of

Ubiquitous and Wearable Computing. By making AI-driven decisions more transparent, we believe this will ultimately enhance user interaction, build trust, and improve the usability of wearable technologies. When users understand the reasoning behind specific functions or malfunctions, they are more likely to trust and effectively engage with these systems. To achieve this, the workshop incorporates a series of structured activities.

- One key objective is to bring together researchers from diverse disciplines, including medicine, computer science, and the social sciences. By fostering collaboration, the workshop provides a platform for networking and the building interdisciplinary partnerships that drive innovation in XAI for ubiquitous and wearable computing.
- Another focus is to discuss recent developments in the field. Participants share the latest research findings, technological advancements, and methodological innovations, ensuring the community remains informed about emerging opportunities and contributions. This exchange of knowledge helps researchers and practitioners stay up to date with evolving landscape of XAI applications.
- Additionally, the workshop identifies and deliberates on open challenges that remain insufficiently addressed. By highlighting gaps, limitations, and unresolved issues in XAI for ubiquitous and wearable computing, it encourages discussions on pressing problems that require further attention. This collaborative, problem-solving approach aims to refine current methodologies and propose novel solutions.
- Finally, the workshop is dedicated to shaping the future of the field by proposing concrete action items for research. Through structured discussions, participants work toward developing a roadmap or set of recommendations to guide future efforts. These insights help direct the community's focus toward impactful and necessary advancements, ensuring continued progress in making XAI more effective, user-friendly, and widely applicable across ubiquitous and wearable technologies.

3 Workshop Structure

Last year's workshop was a great success [8]. It featured two presentation sessions, including seven paper presentations, a keynote by Prof. Tim Miller, and a concluding group discussion on key topics that emerged throughout the day. A post-workshop survey revealed that 100% of respondents were interested in attending future workshops, with particular praise for the group discussion and keynote. This year, we aim to maintain the successful format used last year.

The workshop will include five segments: two keynote presentations, oral presentations for accepted papers, a poster session, a panel discussion, and group discussions. The first keynote will be delivered by Associate Professor Brian Y. Lim from the NUS UbiComp Lab at the National University of Singapore, whose research focuses on developing interpretable, ubiquitous AI by leveraging machine learning and human-computer interaction. The second keynote will be given by Dr. Katharina Weitz from the Fraunhofer Institute for Telecommunications, Heinrich-Hertz-Institut (HHI), Germany. Dr. Weitz has an interdisciplinary background in computer science and



psychology, and her research focuses on human-centered aspects of XAI in application domains ranging from healthcare to disaster risk reduction. Following each keynote, the agenda will include three to five oral presentations by authors of selected workshop papers. These authors will also be encouraged to prepare posters to facilitate deeper discussions and networking during breaks. To broaden the impact of the workshop, we will invite authors of XAI-related papers from the main conference to participate in at least the poster session—that is, these authors will also be invited to prepare posters for the workshop, thereby enriching the variety and depth of content available.

The panel discussion will include four to five experts in the field, including the two keynote speakers. Additional panelists will be determined later, with workshop organizers available as backup panelists and moderators. Based on feedback from previous years, group discussions will form another crucial part of the workshop. Participants will be divided into three to four groups, with each group focusing on challenges in XAI for ubiquitous and wearable computing that emerge throughout the day. The groups will brainstorm possible approaches, potential collaborations, and solutions. At the end, each group will give a short presentation to share their findings.

The workshop will conclude with a summary of the day's discussions and a farewell, encapsulating the insights gained and ideas generated. The tentative schedule is as follows:

- 09:00 - 09:15: Welcome
- 09:15 - 10:00: Keynote 1 + Q&A
- 10:00 - 10:15: Oral presentations - Part 1
- 10:15 - 10:45: Coffee break + Poster tour
- 10:45 - 11:15: Oral presentations -Part 2
- 11:15 - 12:00: Keynote 2 + Q&A
- 12:00 - 12:30: Group discussion summaries
- 12:30 - 12:45: Conclusion and farewell
- 12:45 - 14:00: Lunch break

4 Workshop Submissions

We invite submissions of original research, insightful case studies, and work-in-progress exploring XAI approaches for ubiquitous and wearable computing. Submissions are accepted in two categories:

- (1) **Standard submissions:** Up to 4 pages (including references), submitted through the Precision Conference System (PCS). These submissions will undergo peer review. Accepted standard submissions will be published in the ACM Digital Library and Adjunct Proceedings.

- (2) **Short submissions:** Up to 2 pages (including references), submitted via an alternative platform (e.g., Google Forms). Accepted short submissions will not be published in the ACM Digital Library and Adjunct Proceedings, but will instead be made available on the workshop website.

We anticipate receiving approximately 15–20 paper submissions in total, with an expected workshop attendance of 25–30 participants. The workshop will be open to both registered participants and individuals who have not explicitly registered for our workshop. Pre-workshop preparations include a plan to engage the community and attract high-quality submissions. These efforts include setting up a dedicated workshop website¹, issuing a call for papers, and leveraging social media platforms and mailing lists for widespread dissemination. We will also engage directly with potential attendees through personalized invitations to build momentum and ensure a successful, well-attended workshop.

5 Impact and long-term objectives

The workshop is expected to have a significant impact on the community by fostering a focused discussion on XAI within ubiquitous and wearable computing. Given the absence of similar workshops in recent years at UbiComp, this event will encourage the exchange of ideas, advancements, and solutions specific to XAI in this domain. It has the potential to catalyze a series of workshops at future UbiComp conferences, thereby nurturing a dedicated community of researchers and practitioners committed to advancing XAI in ubiquitous and wearable computing.

As a post-workshop activity, a comprehensive summary report or whitepaper will be released (e.g., on arXiv and shared via email with all workshop participants), encapsulating the discussions, insights, and conclusions from the event. This document will serve as a valuable resource for the broader research community, offering a synthesized overview of the state-of-the-art and future pathways for XAI in ubiquitous and wearable computing. We will also consider publishing the workshop summary as a peer-reviewed article or using it as the foundation for a future scoping review.

6 Organizers' backgrounds

Mor Vered is a senior lecturer at the Department of Data Science and Artificial Intelligence at Monash University. Her research interests lie in the interaction between humans and intelligent agents, where she works to incorporate lessons and inspiration from cognitive science, neuroscience, and biology. As such, she specialises in human-centered XAI, generating explanations grounded in cognitive theories, considering that these explanations need to be consumed by people and therefore taking into account human situation awareness models. She co-organized the previous XAI for U workshop at UbiComp 2024, AAMAS-24/25 Workshops on Rebellious and Disobedient Agents in AI (RaD-AI), and Explainable AI Workshop at IJCAI, 2024/2025.

Teena Hassan is a professor of Computer Science at the Institute for Artificial Intelligence and Autonomous Systems (A2S) at the Bonn-Rhein-Sieg University of Applied Sciences in Germany. She specialises in affective computing and human-robot interaction for diverse assistive scenarios, particularly in the healthcare sector.

¹<https://ubicom-p-xai.github.io>

Her research also includes the application of XAI methods to pain recognition and the development of software architecture for autonomous explanation generation in social robots. She co-organised the previous XAI for U workshop at UbiComp 2024, a workshop titled “Multimodal Co-Construction of Explanations with XAI” at ICMI 2024, and a workshop on empathic human-robot interaction at the European Robotics Forum 2017.

Mandani (Mado) Ntekouli is a postdoctoral researcher in the Department of Advanced Computing Sciences at Maastricht University. Her research interests lie in machine learning and its applications within the field of medicine and biomedical engineering. Her current work focuses on the integration of multi-modal data sources, specifically Ecological Momentary Assessment and sensor data, and the use of XAI techniques to enable early detection of symptom deterioration and support personalized interventions in mental health care.

Sang Won Bae is an assistant professor at Stevens Institute of Technology. She serves as the Director of the CARE AI Lab and is an affiliate faculty member of both the Stevens Institute for Artificial Intelligence and the Semer Center for Healthcare Innovation. Her research focuses on behavior modeling and the development of just-in-time adaptive interventions using mobile and ubiquitous computing, XAI, machine learning, and human-centered AI, with a particular emphasis on healthcare and well-being. She also holds several leadership roles, including Associate Chair for CHI from 2023 to 2026 and Associate Editor for IMWUT from 2023 to 2027.

Martin Gjoreski is an Ambizione Fellow funded by the Swiss National Science Foundation (SNSF). His primary research focus is on AI, specifically in the areas of machine learning, federated learning, and XAI. He applies these concepts to various domains such as ubiquitous computing, affective computing, and digital healthcare. He co-organized the previous XAI for U workshop at UbiComp 2024, UbiTention Workshop at UbiComp 2020, and has served as an Associate Editor for IMWUT since 2023.

Acknowledgments

We acknowledge the financial support for this workshop from:

- Swiss National Science Foundation (SNSF), Project XAI-PAC: Towards Explainable and Private Affective Computing (Project number: 216405).
- Ministerium für Kultur und Wissenschaft (MKW) des Landes Nordrhein-Westfalen (NRW), “Profilbildung 2022” project: Zentrum Assistive Technologien Rhein-Ruhr.

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